

FULL WALKTHROUGH • REVISION 3

# 6-DoF Deblur Build Order

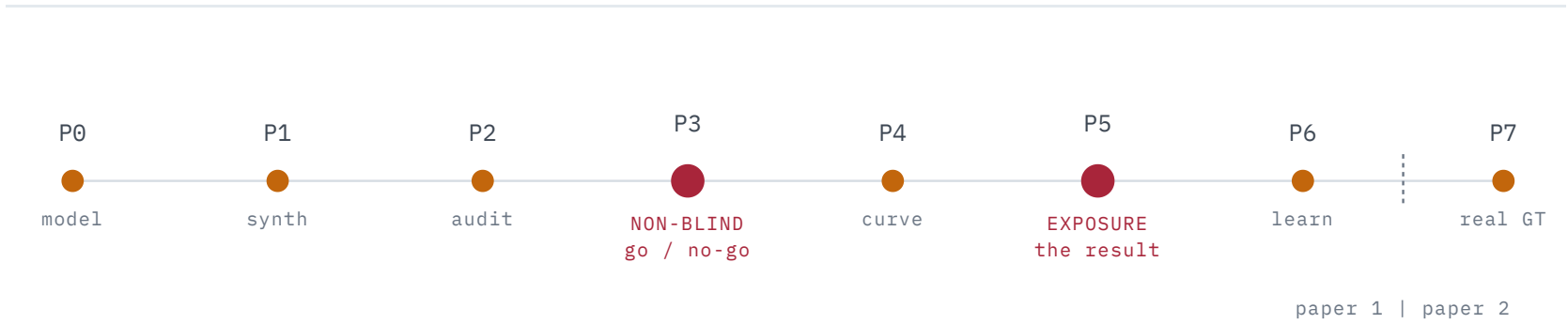
End-to-end research plan: EuRoC as the trajectory and geometry source, synthesis for images, exposure-time estimation as the contribution. Every phase has a gate that can stop it.

## SETTLED — DO NOT RELITIGATE

- **Data.** EuRoC V1/V2 supplies real drone trajectories, hardware-synced IMU, 6-DoF Vicon pose, and a Leica room scan. It does *not* supply sharp images, exposure time, or rolling shutter — and those three are exactly what a renderer sets itself.
- **Never deblur your own ground truth.** Sharp GT comes from synthesis, never from IMU-deblurring EuRoC frames. See the traps below.
- **Global shutter for paper one.** Rolling shutter is a second paper with its own data.
- **Classical before learned.** The optimisation pipeline is the baseline, the debugger, and the data generator.
- **The contribution is exposure time.** Joint estimation of  $\tau$ ,  $t_{\text{offset}}$  and the trajectory, with the identifiability argument for why one image alone cannot do it.

## SHAPE OF THE WORK

### Eight phases, two gates, two papers



PHASES

The walkthrough

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <div>P0</div> <div>WK 1-3</div> | <div>Foundations and data unpacking</div> <div><ul style="list-style-type: none"><li>Unpack V1_01_easy.zip and verify what the docs don't: the point-cloud path (community convention is mav0/pointcloud0/data.ply, unconfirmed) and the state_groundtruth_estimate0 rate.</li><li>Build the loader: Vicon pose + 200 Hz IMU + Leica room scan, all in one coordinate convention. Write the frame-convention test <i>first</i> — this is where silent sign errors live.</li><li>Extend pmbm6: occlusion-aware z-buffered scatter, saturation <math>R(\cdot)</math>, sensor noise, and the rolling-shutter path (built now, used in P7).</li></ul></div> <div><div>GATE</div><div>Adjoint test at <math>10^{-16}</math>. A planted twist recovers within the predicted covariance. Rendered depth from the Leica scan reprojects onto real EuRoC frames with edges aligned.</div></div> |
| <div>D1</div>                   | <div>The synthesiser — your central asset</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

WK 3-6

## The Synthesizer — your central asset

Everything downstream depends on this, so it gets three weeks rather than a weekend.

- **Sharp latents:** frames from V1\_01 / V2\_01 where motion is slowest. These keep EuRoC's real texture, sensor noise and photometrics, which a COCO image does not have.
- **Trajectories:** real Vicor segments of length  $\tau$ , resampled from the 200 Hz IMU. This is the whole point — real drone dynamics, not hand tremor.
- **Depth:** render the Leica scan through the Vicor pose to get a per-frame depth map.
- **Controls:**  $\tau$ ,  $\alpha$ , noise level, saturation, and a virtual camera orientation — so you can render a *nadir* view from the same room scan and compare against forward-facing.

### GATE

Round-trip: plant  $\xi$ , render, re-estimate. Recovery within the covariance predicted from the Fisher matrix. If it isn't, the renderer is wrong, not the estimator.

## P2

WK 6-7

## Information audit

Compute the weighted Fisher matrix using the sharp image's structure tensor and the depth variance, per envelope, on real EuRoC geometry.

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 $\Sigma_p = \sigma^2_{\text{meas}} \cdot \mathbf{I} + \sigma^2_{p,p} \cdot (\tau \cdot \mathbf{A}_p \mathbf{v})(\tau \cdot \mathbf{A}_p \mathbf{v})^T$ 
 $\mathcal{F} = \Sigma_p \mathbf{L}_p^T \Sigma_p^{-1} \mathbf{L}_p$ 
 $\delta \xi_{\text{worst}} = \sigma_{\text{meas}} / \sigma_{\min}(\mathbf{L})$ 
// rotation columns carry no Z —  $\omega$  is immune to depth error

```

### GATE

EuRoC's camera is forward-facing, so

### SURGE SITS ON THE OPTICAL AXIS

— the envelope with no rotational proxy and no  $f$  amplification. Decide here whether to render nadir instead, or to drop metric surge from your claims. Cheap now, expensive later.

**P3**

WK 7-11

**Non-blind deconvolution — the project pivot**

Simplest complete pipeline:  $\xi$  handed over by the gyro, solve for  $u$  only, TV prior, linearised ADMM. Run it three ways — rotation-only, 6-DoF with Leica depth, 6-DoF with a degraded depth map — so depth quality is isolated from model class.

**GATE — GO / NO-GO FOR THE WHOLE PROJECT**

Sweep  $\lambda$  and run the whiteness test. **Does 6-DoF whiten the residual where rotation-only cannot?** If rotation-only already whitens on EuRoC geometry, your premise is weaker than assumed — stop and reconsider before building anything else. Respect the one-sidedness: conclude "inadequate" only if *no*  $\lambda$  whitens.

**P4**

WK 11-13

**Bandlimited SE(3) trajectory**

Take the PSD of EuRoC's own 200 Hz gyro — measure  $f_{\max}$ , don't guess it — and set the curve degree from  $K \geq 2 \cdot f_{\max} \cdot \tau + 1$ . Cubic B-spline on SE(3). Decouple the quadrature count  $T$  from the parameter count  $K$ : use  $T \geq \text{blur extent in pixels}$  so PSFs come out as streaks, not dotted lines.

**GATE**

The curve reproduces the Vicon trajectory to sub-pixel in the  $G = \Sigma L^T L$  metric — not Euclidean on Euler angles.

**P5**

WK 13-19

**Exposure time and timing offset — the contribution**

- **Two routes:** gyro-anchored,  $\tau = \text{extent} / (f \cdot |\omega|)$ ; and the blur-to-flow ratio,  $\tau / T_{\text{frame}} = \text{extent} / \text{flow}$ . At EuRoC's 20 fps the inter-frame flow is  $\sim 45.8$  px against  $\sim 9.2$  px of blur, so the ratio is well conditioned.

- **The stereo trick:** EuRoC's documented flaw is that the two cameras auto-expose independently. Both see the same  $\omega$ , so the ratio of their blur extents is  $\tau_0/\tau_1$ , trajectory cancelled. Cross-check it against the brightness ratio — a validation signal that needs no ground truth, available only because of a dataset defect.
- **Then  $t_{\text{offset}}$ :** 2-D whiteness sweep over  $(\tau, t_{\text{offset}})$ . EuRoC is hardware-triggered so this should come out near zero — which makes it a clean sanity check on your estimator.

**GATE**

EuRoC cannot validate  $\tau$  — it has no exposure metadata in any form. Validate on

**TUM VI**

(per-frame exposure in ns) and

**BSD**

(controlled 1–24 ms), match or beat Korčák & Matas, then *apply* to EuRoC. If it doesn't hold up on TUM VI, the contribution isn't there.

**P6**

WK 19–27

**The learned model**

Architecture follows the maths, not convention:

- Predict **curve coefficients** (~18–30 numbers), not T free poses. The parameterisation is the inductive bias and this is the highest-leverage choice.
- **Condition on the IMU** — feed the gyro segment alongside the image. That breaks the  $\tau$  degeneracy inside the network instead of hoping it learns something provably absent from the image.
- Predict the **residual** from the IMU trajectory, not the absolute. Easier problem, graceful degradation.
- Supervise with  $\min(d(\xi, \xi'), d(\xi, -\xi'))$  under G. Never EMD — it is exactly zero for every permutation, so it cannot teach ordering.
- Drop the roll / pitch-yaw branch split: roll and  $v_z$  share the same  $\propto r$  radial profile.

GATE

Does the learned prior beat BM3D *given that the IMU already supplies the trajectory*? If not, publish the classical pipeline and say so — that is a finding, not a failure.

P7

WK 27-31

Real-data validation, then rolling shutter

Everything to this point is synthetic. Close it out on data you did not generate: **Köhler** (the only real set with GT camera trajectory), **RealBlur**, and **Gen3-DroneFlight** (real drone, sharp GT, mocap, exposure explicitly set to 10/30/50 ms).

Rolling shutter starts here as paper two: simulate for development since you control  $\alpha$ , then validate on **WHU-RSVI** (RS + IMU, readout time published) or **BS-RSCD** (RS + blur + sharp GT). Calibrate  $\alpha$  on hardware by photographing a mains-flicker LED — the banding period gives the line time directly.

GATE

Check separately whether the RS shear is being absorbed into  $u$ . That failure is invisible to PSNR and fatal for photogrammetry.

NAMED NOW, SO THEY CANNOT SURPRISE YOU

Traps

**Circular ground truth** If you deblur EuRoC with your model and call the output sharp GT, you are measuring how well your method reproduces itself. Every systematic error — wrong  $\tau$ , missing  $v_z$ , unmodelled parallax — becomes permanently undetectable, and a network trained on it learns your errors as physics. IMU-deblurred frames are acceptable as *pseudo-labels for pretraining* only, with final evaluation on real pairs.

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**V1\_01\_easy orientation GT** Known to be wrong. OpenVINS ships a corrected version from their `vicon2gt` tool. The images are fine — only the rotation ground truth is bad, and rotation is what you're estimating.

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**Vicon has no device timestamps** Officially acknowledged. Sync between sensor data and motion GT is limited by separate recording systems. Hand-eye and time alignment is a real preprocessing step, not a formality.

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**Coordinate conventions differ** Ground truth, IMU and camera frames use different axis conventions — evo warns about this explicitly. In interaction-matrix work a sign error is silent and looks like a modelling failure.

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**Machine Hall is position-only** Leica laser tracker, 3-DoF, and no room scan. Use V1/V2 for anything trajectory-related. MH sequences also open with the MAV being picked up and set down — skip that segment.

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**$v_z$  conditioning** Forward flight gives radial displacement with no  $f$  amplification:  $\sim 1.2$  px against  $\sim 9$  px for yaw. The most-used axis is the least-encoded one, and it is confusable with defocus. Don't promise metric forward velocity.

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**Soft depth boundaries** Whenever you swap the Leica scan for a learned depth map, occlusion boundaries go soft where true depth is discontinuous. With a  $z$ -buffered scatter, that is where artefacts appear — and it is the one place existing theory gives no cover.

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## REFERENCE

### EuRoC numbers you'll need

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| QUANTITY              | VALUE                                           | CONSEQUENCE                                       |
|-----------------------|-------------------------------------------------|---------------------------------------------------|
| cam0 intrinsics       | $f_x$ 458.65, $f_y$ 457.30, $c$ 367.22 / 248.38 | HFOV $\approx 79^\circ$                           |
| Distortion (radtan)   | -0.28341, 0.07396, 1.94e-4, 1.76e-5             | undistort before eq. (23)                         |
| Stereo baseline       | 110.1 mm                                        | metric depth available                            |
| Frame rate            | 20 fps $\rightarrow$ 50 ms period               | blur-to-flow ratio $\approx 0.2$                  |
| Gyro noise density    | 1.6968e-4 rad/s/ $\sqrt{\text{Hz}}$             | 0.008 px over 10 ms                               |
| Gyro random walk      | 1.9393e-5                                       | 0.046 px at 0.01 rad/s bias                       |
| Accel noise density   | 2.0e-3 m/s <sup>2</sup> / $\sqrt{\text{Hz}}$    | negligible over $\tau$                            |
| Crossover depth $Z^*$ | $f \cdot v \cdot \tau = 9.16$ m                 | below it, depth is needed <i>and</i> measurable   |
| Depth tolerance       | 11% at 2 m, 27% at 5 m                          | $\delta Z/Z < 0.5 \cdot Z/(f \cdot v \cdot \tau)$ |

The gyro rows are the ones that matter most: the IMU contributes three orders of magnitude less error than your 0.5 px kernel noise floor, so gyro-anchoring — the load-bearing assumption of the whole plan — is comfortably valid on this data.

Companion material: MATH\_FOUNDATION.md, the 6-DoF addendum (§13–§23), and pmbm6/. Start with P0's frame-convention test and unzipping V1\_01 — a day's work, and it settles two things the documentation cannot.

